Business Requirements Document (BRD)

Meta Ad Performance Analysis

Platforms: Facebook & Instagram | Tool: Power BI | Analysis Period: July

1. Business Objective

The business required a performance tracking dashboard for paid advertising campaigns running on Facebook and Instagram. The dashboard needed to provide full visibility into campaign reach, engagement, conversions, and budget utilization — and support platform-level filtering so the marketing team could compare Facebook vs. Instagram side by side.

This enabled the marketing team to:

- Identify the most effective platform (Facebook vs. Instagram) across all KPIs
- Track campaign ROI and optimize budget allocation across campaign tiers
- Understand audience engagement patterns by gender, age, country, and time

How the Dashboard Answered This

> **Dashboard Answer:** A single-page interactive Power BI dashboard was built with a platform slicer (Facebook / Instagram) that filters all 10+ visuals simultaneously. Facebook delivered 216K impressions and 1.3K purchases. Instagram delivered 123.8K impressions and 708 purchases — combined total of 339.8K impressions across a 2.54M total budget.

2. Scope of the Report

In Scope

- Campaigns running on Facebook and Instagram only
- Paid ad events only: Impressions, Clicks, Shares, Comments, Purchases
- Platform-level filtering (Facebook vs. Instagram toggle on dashboard)
- Demographic analysis: gender, age group, country, interests
- Time-based analysis: hourly, daily (calendar), weekly trends
- Campaign budget vs. performance tiering: Hidden Gem, High Performer, Low Priority, Overspending

Out of Scope

- Other Meta platforms: Messenger, Audience Network
- Organic engagement (non-paid interactions)
- Post-purchase customer journey / CLV data

3. KPIs & Definitions

The following 12 KPIs were defined, agreed with stakeholders, and implemented as DAX measures in Power BI. All KPIs are dynamic — they update in real time based on the platform slicer and campaign/interests filters.

KPI	Definition	Formula	Example Use
Impressions	Total times ads were displayed	COUNT(event_type = Impression)	Measure reach
Clicks	Number of times users clicked ads	COUNT(event_type = Click)	Measure engagement intent
Shares	Number of times ads were shared	COUNT(event_type = Share)	Viral engagement
Comments	Number of user comments on ads	COUNT(event_type = Comment)	Sentiment & feedback
Purchases	Number of purchases made after seeing ads	COUNT(event_type = Purchase)	Conversions
Engagements	Total interactions (Clicks + Shares + Comments)	Clicks + Shares + Comments	Engagement volume
CTR	% of impressions that resulted in clicks	(Clicks ÷ Impressions) × 100	Ad effectiveness
Engagement Rate	% of impressions that resulted in engagements	(Engagements ÷ Impressions) × 100	Overall ad appeal
Conversion Rate	% of clicks that resulted in purchases	(Purchases ÷ Clicks) × 100	Funnel efficiency
Purchase Rate	% of impressions that resulted in purchases	(Purchases ÷ Impressions) × 100	Conversion from reach
Total Budget	Total spend allocated to all campaigns	SUM(campaigns.total_budget)	Cost analysis
Avg Budget/Campaign	Average budget allocation per campaign	Total Budget ÷ Campaign Count	Budget distribution

> **Insight:** *All KPIs are powered by a dynamic measure parameter slicer — selecting any KPI instantly updates every visual on the dashboard, including the map, gender donut, age bar chart, and ad-type matrix.*

4. Platform-Level KPI Results

The dashboard supports a platform toggle (Facebook / Instagram) that filters all visuals independently. The table below shows the actual values recorded in the dashboard for each platform:

KPI	Facebook	Instagram	Key Difference
Impressions	216,000	123,800	Facebook delivers 75% more reach
Clicks	25,400	14,700	Facebook generates ~73% more clicks
Purchases	1,300	708	Facebook drives ~2x more conversions
CTR	11.76%	11.86%	Instagram edges ahead on CTR
Conversion Rate	5.21%	4.82%	Facebook converts clicks more efficiently
Engagement Rate	13.56%	13.60%	Instagram marginally stronger on engagement
Purchase Rate	0.61%	0.57%	Facebook more efficient impression-to-sale
Total Budget	2.54M	2.54M	Shared budget across both platforms
Avg Budget/Camp.	50.72K	50.72K	Same campaign structure on both platforms

> **Insight:** *Facebook wins on volume and conversion efficiency. Instagram wins on CTR and engagement rate. Both platforms have a sharp funnel drop-off at the purchase stage — confirming this is a landing page / offer problem, not a platform-specific creative issue.*

5. Dashboard Charts & What They Answered

Each chart was built according to the original requirements and answered a specific business question. Below is a chart-by-chart breakdown of the visual type, the business question it addressed, and what the data actually showed.

Chart 1: Purchases by Gender — Donut Chart

Visual Type: Donut chart with dynamic metric parameter

Business Question: Which gender segment contributes most to the selected KPI?

What the dashboard showed:

Segment	Facebook Purchases	Instagram Purchases	Interpretation
Female	210 (43%)	80 (33%)	Dominant on Facebook; significant on Instagram
All / Unspecified	152 (33%)	97 (40%)	Largest single group on Instagram
Male	93 (20%)	64 (27%)	More balanced gender mix on Instagram

> **Insight:** *Facebook audiences are strongly female-skewed (43%). Instagram shows a more balanced gender split. Female-targeted creatives should be prioritized on Facebook; Instagram can support broader creative approaches.*

Chart 2: Weekly Purchases by Age — Bar Chart

Visual Type: Bar chart with user_age on the X-axis and selected metric on the Y-axis

Business Question: Which age group is most responsive to campaigns?

What the dashboard showed:

- Peak engagement and purchases concentrated in the 18–30 age range on both platforms
- Engagement drops sharply after age 35 across Facebook and Instagram
- Users aged 50+ show minimal interaction with ads

> **Insight:** *The primary audience is young adults aged 18–30. All targeting parameters, creative design, and messaging tone should be optimized for this age cohort on both platforms.*

Chart 3: Purchases by Country — World Map

Visual Type: Bubble map with bubble size representing the selected metric by country

Business Question: Which countries drive the most reach and engagement?

What the dashboard showed:

- US, India, Brazil, Germany, and UK are the top 5 engaged countries across both platforms
- India and Brazil: highest volume of impressions and engagement
- Germany and UK: smaller volume but higher purchasing power, suggesting better conversion potential

> **Insight:** *Run high-frequency, broad campaigns in India & Brazil for volume. Run premium, conversion-focused campaigns in Germany & UK where purchasing intent is stronger.*

Chart 4: Analysis by Month — Calendar Heat Map

Visual Type: Calendar heat map with darker shades indicating higher activity

Business Question: Are there seasonal or day-of-month patterns in campaign activity?

What the dashboard showed:

- Engagement spikes visible around the 19th–21st and 25th–27th of July
- These clusters likely correspond to scheduled promotions or campaign launches
- Weekday activity generally outperforms weekend activity across the month

> **Insight:** *Event-based campaigns drive measurable engagement spikes. A formalized bi-weekly promotional cadence (mid-month and end-of-month) could systematically replicate these peaks.*

Chart 5: Weekly Purchases Trend — Stacked Column by Ad Type

Visual Type: 100% stacked column chart, weeks on X-axis, ad types as color stacks

Business Question: How do different ad formats contribute to performance week by week?

What the dashboard showed:

- All four ad types maintained relatively consistent weekly contribution shares
- Stories consistently occupied the largest stack share throughout the period
- No single week showed a dramatic format shift — engagement was structurally stable

> **Insight:** *Stories' dominant weekly share on Facebook aligns with its highest absolute impressions (24,219). On Instagram, the weekly stacks are more evenly distributed across all four formats.*

Chart 6: Hourly Purchases Trend — Area Chart

Visual Type: Area / line chart with hour of day (0–23) on the X-axis

Business Question: At what times of day do users engage most with ads?

What the dashboard showed:

- Engagement peaks between 15:00–20:00 on both Facebook and Instagram
- Lowest activity between 0:00–5:00 (early morning hours)
- The hourly pattern shows multiple peaks across the day — not just one prime-time window

> **Insight:** *Budget weighting and bid strategies should be shifted toward the 15:00–20:00 window on both platforms. Ad spend during 0:00–5:00 should be minimized or paused to reduce wasted impressions.*

Chart 7: Analysis by Ad Type — Matrix Table

Visual Type: Conditional-formatted matrix table (rows: ad types; columns: KPI metrics)

Business Question: Which ad format delivers the best performance across all KPIs, and does this differ by platform?

Facebook Ad Type Results:

Ad Type	Impressions	Purchases	Clicks	CTR	Engagements	Purchase Rate
Stories	24,219	156	2,885	11.91%	3,352	0.64%
Image	17,233	111	2,042	11.85%	2,337	0.64%
Video	15,542	93	1,919	12.35%	2,222	0.60%
Carousel	16,217	95	1,904	11.74%	2,185	0.59%

Instagram Ad Type Results:

Ad Type	Impressions	Purchases	Clicks	CTR	Engagements	Purchase Rate
Carousel	13,223	90	1,536	11.62%	1,765	0.68%
Image	12,655	68	1,510	11.93%	1,731	0.54%
Stories	12,683	71	1,490	11.75%	1,713	0.56%
Video	3,561	12	412	11.57%	476	0.34%

> **Insight:** *Video dominates on Facebook (highest CTR: 12.35%) but dramatically underperforms on Instagram (lowest CTR: 11.57%, lowest Purchase Rate: 0.34%). On Instagram, Carousel achieves the highest Purchase Rate (0.68%). Platform-specific creative strategies are essential — do not reuse the same format across both platforms.*

Chart 8: Budget vs. Performance — Scatter Plot

Visual Type: Scatter plot with bubble sizing; X-axis: total_budget; Y-axis: Purchases; Color: Campaign Tier

Business Question: Are campaigns spending budget efficiently relative to their purchase output?

What the dashboard showed:

- Several 'Overspending' campaigns (high budget, low purchases) visible in the 60K–100K budget range
- Multiple 'Hidden Gem' campaigns deliver strong purchase rates at low cost
- 'High Performer' campaigns clustered in the 40K–80K budget range with above-average purchases

> **Insight:** *Immediately audit and reduce budgets on 'Overspending' campaigns. Reallocate that spend toward 'Hidden Gem' and 'High Performer' campaigns to lift overall ROAS without increasing total budget.*

Chart 9: Conversion Funnel

Visual Type: Horizontal funnel bar chart — Impressions → Clicks → Purchases

Business Question: Where in the funnel is the biggest drop-off occurring?

What the dashboard showed (Facebook):

Funnel Stage	Volume	% of Impressions	Drop-off
Impressions	73,200 (7.32E+04)	100%	—
Clicks	8,750 (8.75E+03)	~12%	88% drop from impressions
Purchases	455 (4.55E+02)	~0.6%	95% drop from clicks

> **Insight:** *The critical drop-off happens between clicks and purchases — not between impressions and clicks. Users ARE clicking the ads (CTR 11.76%), but ~95% of those clickers do not purchase. This is a post-click problem: landing page experience, offer relevance, or checkout friction — not an ad creative problem.*

6. Business Questions Answered — Summary

The table below maps each original business objective from the BRD to the specific dashboard element that answered it and the key finding discovered.

Business Objective	Dashboard Element	Finding
Identify most effective platform	Platform slicer + all charts	Facebook: higher volume & conversion. Instagram: higher CTR & engagement rate.
Track campaign ROI	Budget vs. Performance scatter	Overspending campaigns identified. Hidden Gems and High Performers flagged for reinvestment.
Understand audience engagement	Gender donut + Age bar chart	Core audience: females aged 18–30. Facebook skews female (43%); Instagram more balanced.
Geographic reach analysis	Country map	High volume: India & Brazil. High value: Germany & UK. Separate strategies recommended.
Identify best ad formats	Ad type matrix table	Video best on Facebook. Carousel/Image best on Instagram. Do NOT cross-apply formats.
Optimize ad scheduling	Hourly area chart	Peak hours: 15:00–20:00. Reduce/pause spend between 0:00–5:00.
Detect seasonal trends	Calendar heat map	Bi-weekly engagement spikes (19th–21st, 25th–27th) confirm event-based campaign value.
Measure funnel efficiency	Conversion funnel visual	CTR is strong (11.76–11.86%). Purchase rate is weak (0.57–0.61%). Problem is post-click.